

PIPE SUPPORTS AND RESTRAINTS

Providing sufficient pipe wall thickness and installing proper supports are two of the most important elements in ensuring structural integrity of the piping system. An adequate wall thickness is needed to contain the process fluid, and a proper support system is required for holding the pipe in place. Pipe supports are generally referred to as devices used in supporting the weight of the piping. The weight includes that of the pipe proper, the content the pipe carries, and the pipe covering, such as insulation, refractory, lining, and snow.

In addition to supports, a piping system may also need restraints to control movement, resist occasional loads such as wind and earthquake, protect sensitive equipment, increase stiffness, reduce vibration, and so forth. The purpose of using restraints is to restrict the movement of the piping in certain directions. Both restraints and supports for weight are collectively referred to as pipe supports.

6.1 DEVICE TERMINOLOGY AND BASIC FUNCTIONS

A pipe support and restraint system involves many different types of hardware components and arrangements to serve the different functions needed. In a computerized stress analysis environment, it is necessary to have a common terminology so that design engineers, computer specialists, and fabricators can all communicate with a common language. The following are some of the pipe support terminologies used by the piping community. Most of these terms originated from Kellogg's book [1], although some of the original meanings have been slightly refined. These terminologies are explained with their associated basic functions. The methods of describing these functions to the computer are also discussed.

Before discussing each device, it is important to remember that every point of the piping system is associated with six degrees of freedom: three directions of translation and three directions of rotation. Without restriction, the pipe can move in the x , y , and z directions, and can also rotate about the x , y , and z axes. The supports and restraints normally restrict the motion at one or more of the degrees of freedom. The effectiveness of the restriction to each direction depends on the stiffness of the support structure in that particular direction. This stiffness is commonly called the support spring constant or spring rate. Theoretically, a rigid support or restraint has an infinitely large spring rate. In actual applications, a sufficiently high spring rate may be used instead to avoid mathematical difficulties.

- Anchor

An anchor fixes all six degrees of freedom. This is the most fundamental support in piping stress analysis. Most of the basic flexibility analyses are done by assuming that both ends of the piping system are anchored. Figure 6.1 shows some of the common arrangements of piping anchors. Equipment connections are normally considered anchors. However, other than equipment connections, there are very few real anchors that exist in a piping system. The places most likely to have an anchor placed are at the ends of an expansion joint and at the ends of a long pipeline.

A theoretical anchor does not allow any pipe displacement or rotation at the anchor point. This is because the anchor is assumed to have an infinite stiffness in all six degrees of freedom.

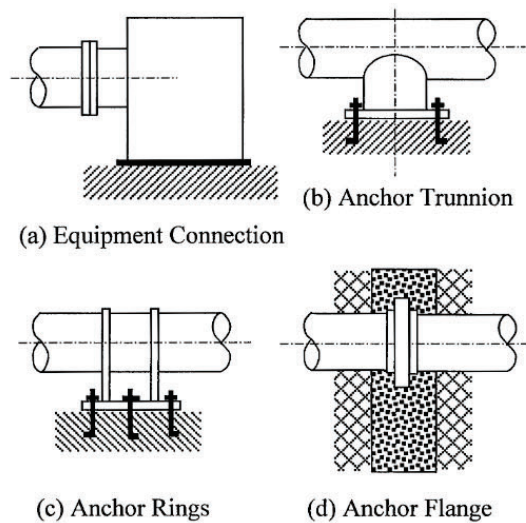


FIG. 6.1
PIPING ANCHORS

However, some of the equipment connections, such as vessels and tanks, do have significant flexibility in certain degrees of freedom. These connection flexibilities are handled either with a flexible connection attached to the anchor or by considering the anchor as six directional restraints, each with its own stiffness. In addition to the fixation, the equipment connection also normally exerts some displacements to the piping, due to the thermal expansion of the equipment itself. These displacements also need to be applied in the analysis. The anchor shown in Fig. 6.1(d) is normally used to limit the large end displacement of a long underground pipeline. Because it is very difficult to achieve a complete fixation under some poor soil conditions, the anchor block is occasionally designed to allow it to plow through the soil somewhat. This type of anchor offers no complete fixation. It only provides drag and prevents the pipe from moving too much, and is thus called a drag anchor.

• Supports

A support device is used to sustain a portion of the weight of the piping system and other superimposed vertical loads. However, a support is specifically referred to as the device acting from underneath the pipe, in contrast to the hanger that is working from above the pipe. As noted before, “pipe support” is also used as the generic term for all supports and restraints combined.

Figure 6.2 shows some general types of rigid hangers and supports. The hanger shown in Figure 6.2(a) will be defined and discussed separately. Support types are distinguished mainly by the pipe shoes they use. The most direct and economic supporting scheme is to rest the pipe directly on the support structure as in Figure 6.2(b). When the pipe needs to be insulated, a shoe is generally required. Figure 6.2(c) shows a shoe made from a piece of inverted-T steel. This T shoe has a single loading line placed at the weakest mid-shell location of the pipe. Detail (c) is used only for smaller pipes, up to 10 in. (250 mm) in size. For larger pipes, say between 12 in. (300 mm) through 24 in. (600 mm), the H shoe as shown in (d) can be used. The H shoe, taken from a wide flange I-shape steel, divides the load into two lines placed near the sides of the pipe shell. The

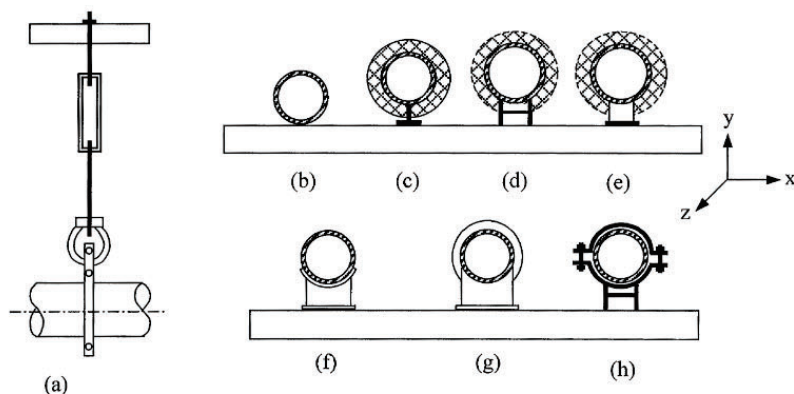


FIG. 6.2
RIGID HANGER AND SUPPORTS

side of the pipe shell can take a much larger load than the center portion of the shell. When the pipe temperature exceeds 750 °F (400 °C), the temperature gradient stress on the T and H shoes can reach excessive levels, thereby requiring the use of a trunnion shoe (e) or a clamped shoe (h). The saddle (f) and ring girder (g) are mainly for pipe sizes larger than 24 in. (600 mm).

In piping stress analysis, the hanger and supports given in Fig. 6.2 are all classified as stops in the y direction. Because these supports only stop the piping from moving downward, a more elaborate analysis will classify these as stops in the minus ($-$) y direction. The y direction is used by many computer software packages to represent the vertical direction pointing upward — that is, the direction directly opposing the direction of gravity.

- Hanger

Similar to the support, the hanger is used specifically to sustain a portion of weight of the piping system and superimposed vertical loads. However, a hanger sustains the piping weight from above, and its load is always in tension. Due to this tensile nature, a slender rod can be used without the concern of buckling. A hanger normally refers to a rigid hanger. Although a hanger may have considerable flexibility in the loading direction, it is still generally much stiffer than the piping system in the direction supported.

A hanger is suspended from a structure with the hanger rod pivoting at a fixed point as shown in Fig. 6.3. When the pipe moves due to thermal expansion or other forces, the hanger rod slants creating a horizontal resistance. It can also lift up the piping somewhat. These types of secondary effects are normally not implemented in typical analysis methods or in computer software. However, if the slanting angle is limited to no more than 4 deg., then the horizontal resistance can be ignored. Figure 6.3 shows that at a 4 deg. slanting, the horizontal resistance is about 70 units per 1000 units of support force. To reduce the effect of this horizontal resistance to a minimum, a hanger is normally installed with some initial slanting so that it becomes vertical at the operating condition. When the headroom of the piping is insufficient to achieve this less than 4 deg. slant angle, a trapeze hanger may be used.

- Restraint

Any device that prevents, resists, or limits the movement of the piping is called a restraint. Restraints refer generally to devices other than the weight sustaining devices, or supports defined

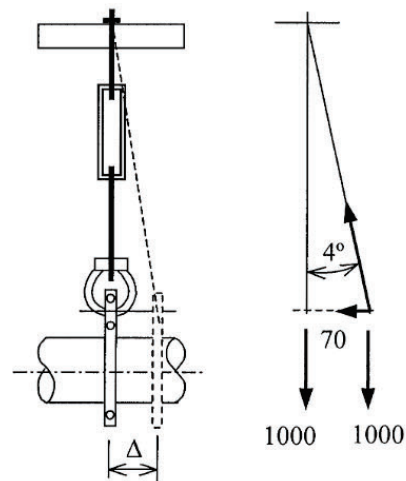


FIG. 6.3
LATERAL FORCE CREATED BY DISPLACEMENT

above. Figure 6.4 shows some of the more commonly used restraints. There is no limit as to how many directional restraints can be installed at one point. The guides and stops shown in the figure are all combined with a vertical support. An anchor can also be considered as the combination of six directional restraints.

- Strut

A directional restraint consists of a compressive column with two rotational joints. A strut is subjected to both tensile and compressive loads. It is used when there is no suitable support structure nearby. It can also be used to reduce the restraint friction force. A strut can be conveniently attached to a nearby structure in a generally skewed direction.

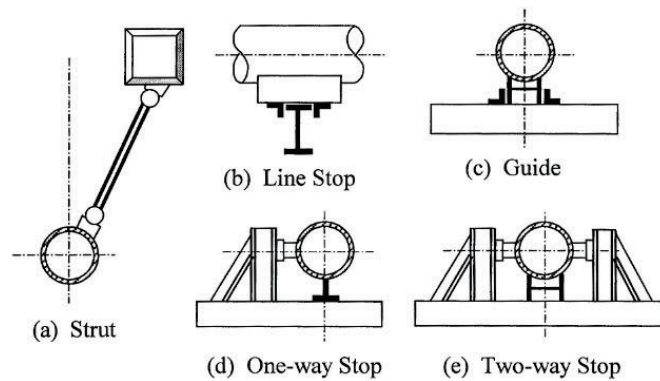


FIG. 6.4
DIRECTIONAL RESTRAINTS

- Stop

This is a device that stops the piping movement in one or more translation directions. Meanwhile, a *line stop* specifically refers to the device that stops the pipe in the axial direction. Unless specified otherwise, a stop acts in both plus and minus directions (also known as double-acting).

- Guide

The guide stops the pipe from moving in the lateral direction. The guide is double-acting in nature. For a long straight segment of piping, guides are generally provided at every other support span. (This term was used by the Kellogg book for rotational restraint.)

- Space maintenance stop

The stop details given in Fig. 6.4(b) and (c) are mainly for space maintenance. They hold the piping in the proper space and resist moderate occasional forces. For large stop forces, more solid constructions, such as the ones shown in Fig. 6.4(d) and (e), should be used.

- One-way stop

This is a directional stop that stops the pipe in either the plus or the minus direction. The pipe is free to move in the opposite direction.

- Double-acting

A restraint that is active in both plus and minus directions. This requires the stop members to be welded on both sides of the support shoe as in Fig. 6.4(b), (c), and (e).

- Single-acting

The device is active only in either the plus or minus direction. If one of the stop members in Fig. 6.4(b) and (c) is removed, the stop becomes single-acting. A one-way stop is single-acting.

- Limit stop

A stop that is active only after the pipe has moved a certain amount. The allowable displacement is controlled by the gap between the pipe and the stop. It should be noted that most stops and guides are installed with a 1/16-in. (1.5 mm) construction gap to prevent binding between the pipe and the stop. Besides those located near equipment, the construction gaps are normally ignored in the analysis.

- Brace

This device is similar to a strut, but is used specifically in resisting occasional loads and reducing vibrations. A brace is commonly attached with a pre-compressed spring or friction unit. It affords a pre-set amount of initial resistance, after which a constant amount or sloping amount of resistance persists. The device does not need any initial movement to activate the stopping action. Figure 6.5 shows a brace with a pre-compressed spring. Detail (a) shows the force-displacement relation, and detail (b) shows the modeling technique. The brace can be modeled as the combination of one linear spring and one elastic-plastic non-linear restraint. A limit stop can also be superimposed to limit the maximum travel.

- Resting or sliding support

A device that provides support from beneath the piping but offers no resistance other than frictional to horizontal motion. A *resting support* is a single-acting device that stops the piping from moving only in the downward direction.

- Rigid (solid) support

A support or restraint that consists of solid structural members only. Although the support structure may possess some inherent flexibility, it is still much stiffer than the piping in the direction of support.

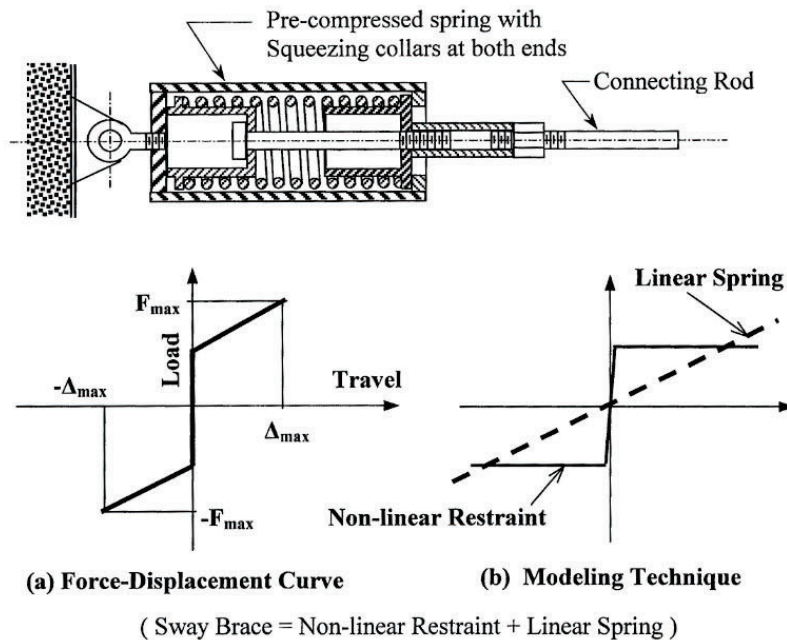


FIG. 6.5
PRE-COMPRESSED SPRING SWAY BRACE

- Spring (resilient) support

A support that includes a very flexible member such as a spring. A *variable spring support* is much more flexible than the piping in the direction supported. It is called a variable support because the magnitude of its supporting load changes as the pipe moves.

- Constant-effort support

A support that applies a relatively constant support force to the piping regardless of the pipe movement (e.g., compensating spring or counterweight device).

- Snubber

A device that resists shock loads such as earthquake and water hammer, but does not resist the slow moving thermal expansion movement. Snubbers can be classified as hydraulic or mechanical as shown in Fig. 6.6. With hydraulic snubbers, the restraint movement pushes a piston inside a fluid filled cylinder. The cylinder has a passage to the reservoir. Inside the passage, there is a spring loaded check valve that allows the fluid to flow through gradually but shuts off the flow when the fluid velocity reaches a threshold limit. The mechanical snubber, on the other hand, uses a ball screw and ball nut assembly to convert the translational restraint movement to rotational movement. When the rotational acceleration reaches a certain level, the rotation of the torque transfer drum is bound still by the capstan spring. Either type of snubber requires a small initial movement before the snubbing action is activated. Therefore, it is not effective in reducing steady-state small amplitude vibrations. In piping stress analyses, the snubber is active for dynamic loads, but not for static loads.

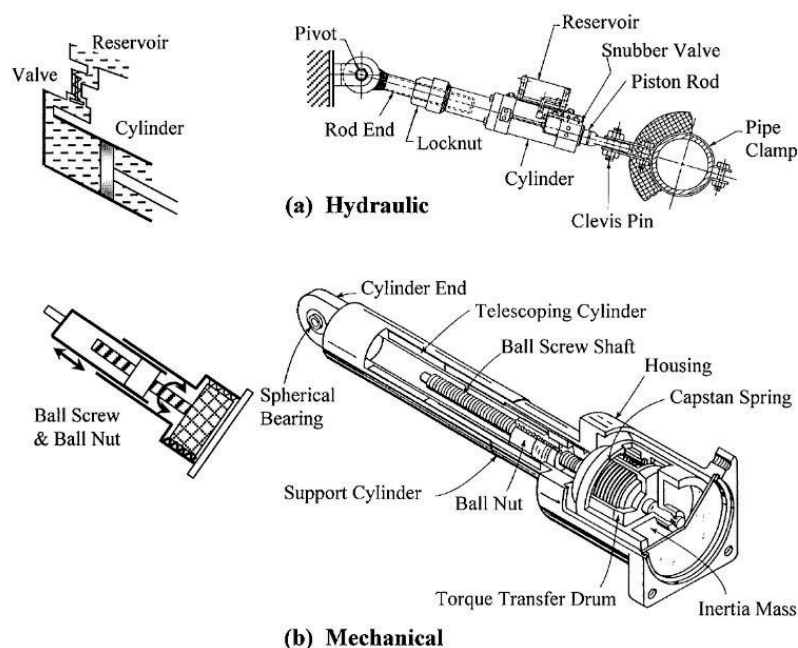


FIG. 6.6
SNUBBERS

- Rotational restraint

This is a device that prevents the pipe from rotating about one specific axis. It is sometimes referred to as a *moment restraint*. The rotation can be prevented with a specially designed device or by two coupled parallel stops. If it is done by two parallel stops, it is also modeled as two stops in the analysis.

6.2 SUPPORT SPACING

Adequacy of the support is determined by the sustained stress limitation. If the sustained stress meets the code requirement, then the support system is considered acceptable. This criterion is what computerized stress analyses are based on. If the sustained stress is within the allowable limits, then it does not matter how often the piping is supported. However, there are other concerns that require the setting of the allowable support spans for horizontal lines. These concerns are discussed in the following subsections.

1. Layout of Support Steels and Sleepers

These support locations are normally determined before the piping system is analyzed for acceptable stress. Layout designers will need allowable spans that ensure safe stress to work with.

2. Limit Pipe Displacements

The displacement of the pipe needs to be limited for better appearance and to prevent excessive pooketing. The displacement limitation is also required to avoid interference between pipes. Although

excessive displacements will be discovered during stress calculations, it is better to limit them when the piping is being routed.

3. Avoid Support Rotations

Excessive pipe rotations twist support shoes making the supports ineffective. In general, limiting the sagging displacements can prevent excessive rotations.

4. Provide Rigidity

A certain amount of piping system rigidity is needed to reduce undesirable vibration. This, in part, is related to limiting the displacements.

The allowable support spans are determined by the stress and displacement criteria. These can be evaluated by the semi-fixed beam approach as discussed in Section 2.7. Based on the allowable bending stress criterion, the spacing is limited to

$$L_1 = \sqrt{\frac{10ZS}{w}} \quad (6.1)$$

where S is the design weight bending stress (B31.1 [2] uses $S = 2300$ psi (15.86 MPa), whereas CEN [3] uses $S = 10$ MPa (1450 psi), and Kellogg [1] uses $S = 2750$ psi (18.96 MPa)); Z is the section modulus of the pipe cross-section; and w is the weight force per unit length of pipe.

Based on the allowable sagging criterion, the spacing is limited to

$$L_2 = \sqrt[4]{\frac{128EI\Delta}{w}} \quad (6.2)$$

Δ is the design sagging displacement. B31.1 uses $\Delta = 0.1$ in. (2.5 mm) for power plants, whereas Kellogg suggests a Δ of 0.5 in. to 1.0 in. (12.5 mm to 25 mm) for process plants.

The allowable span, L_s , is therefore taken as the smaller of L_1 and L_2 . Table 6.1 shows the suggested pipe support spacing by B31.1 for power plants. A similar table can be constructed accordingly for each type of service with different allowable stress and displacement criteria.

The suggested pipe support spacing is applicable only for uniform pipe without any attached concentrated weight, such as a valve or flange. It is not applicable for the overhanging span either. For overhanging spans, the quarter circle approach as shown in Fig. 6.7 can be used. The curve shown is a quadrant of a circle having a radius equal to the allowable spacing. The two coordinates of the point

TABLE 6.1
SUGGESTED PIPE SUPPORT SPACING
FOR POWER PIPING [2]

Nominal pipe sizes		Suggested maximum span, L_s			
		Liquid service		Gas service	
		ft	m	ft	m
1	25	7	2.1	9	2.7
2	50	10	3.0	13	4.0
3	80	12	3.7	15	4.6
4	100	14	4.3	17	5.2
6	150	17	5.2	21	6.4
8	200	19	5.8	24	7.3
12	300	23	7.0	30	9.1
16	400	27	8.2	35	10.7
20	500	30	9.1	39	11.9
24	600	32	9.8	42	12.8

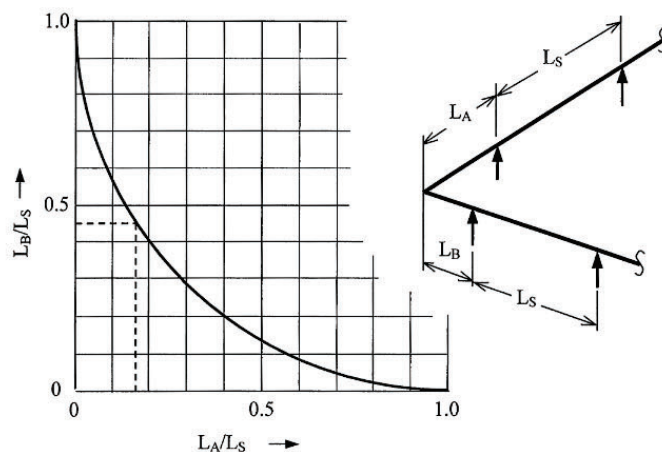


FIG. 6.7
OVERHANGING PIPING SPANS

on the curve determine the two allowable overhanging lengths. For instance, if $L_B = 0.45L_S$, then the allowable L_A is $0.16L_S$.

6.3 ANALYSIS OF PIPING SYSTEMS RESTING ON SUPPORTS

The most common and economical approach in dealing with the countless piping in a process plant or power plant is to rest the piping on pipe racks or other support structures [4]. The piping is supported either directly on the pipe wall or through pipe shoes. These types of supports are generally called resting supports. These supports are single-acting, because they only stop the pipe from moving downward but allow the pipe to move up freely. Due to the nature of this non-linearity, exact solutions are not expected for piping that goes through various temperature cycles. Therefore, three major schools of thought have developed in the pipe stress software community with regard to resting supports and temperature cycles. The analysis results and the qualities of the system designed differ considerably among the methods used.

A simple example shall be used to demonstrate the merits and pitfalls of some analysis approaches designed to satisfy the code requirements and philosophy. Figure 6.8 shows a typical piping system resting on the support structure. The piping has one end connected to a process tower and the other end connected to another process equipment. The system is supported at three locations. As the temperature of the process fluid rises, the tower grows upward and the pipe expands. With the tower connection moving up gradually, the piping system also goes through the following sequence of changes:

- With a small tower movement, the piping is held down on all supports by the weight of the piping including fluid and attachments. Some thermal expansion (displacement) stress is generated, but the weight stress remains the same as in the cold condition.
- As the temperature rises gradually and the tower movement increases somewhat, the piping will lift from the first support (support 20). A further increase of the tower movement will lift the pipe off support 30, thus making a large portion of the piping unsupported. This substantially increases the sustained weight stress.
- As the system reaches the maximum operating temperature, the tower connection moves up some more, but the pipe is still being held down at support 40. The expansion stress increases

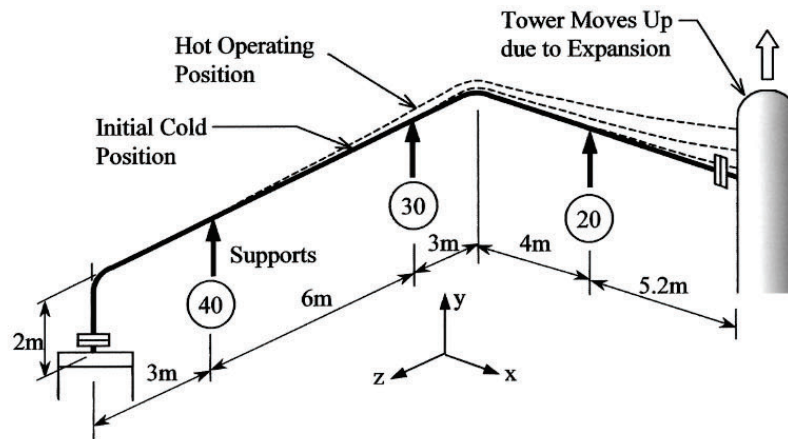


FIG. 6.8
PIPING RESTING ON SUPPORTS

due to the higher temperature and larger tower movement. However, the sustained weight stress remains unchanged as no additional supporting location is lifting off its support.

- When this maximum operating temperature is maintained for a certain period, the thermal expansion stress (displacement stress) will be relaxed somewhat. The amount of relaxation depends on stress level and operating temperature. However, the sustained weight plus pressure stress remains the same.
- When the plant cools down, the piping moves back on to its supports. This reduces the weight stress to its initial cold condition weight stress. The system, however, generates some reverse thermal expansion stress due to relaxation at operating temperature.
- If considerable yielding or creep occurs at hot condition, the pipe may return to the support point before the temperature reaches the ambient temperature. A continued cooling down to ambient temperature will cause high thermal stresses and loads due to stoppage by the support that prevents the pipe from moving further down. Generally, this will cause unpredictably high loads and stresses at the tower connection.
- In the next operating cycle, the weight stress goes back to the hot condition stress sustained, but the expansion stress will be reduced to a level corresponding to the relaxed state.

Three main approaches have been adopted by computer software packages in dealing with systems like the one shown in Fig. 6.8.

1. *General straightforward approach.* This is an approach commonly adopted by general-purpose finite element programs. In this approach, sustained stress and expansion stress are calculated separately without checking the situation influenced by the other load. Sustained stress is calculated by considering only the weight and pressure loads at ambient state. All supports are considered active because no temperature and support displacement is involved. The expansion stress range is calculated only with the temperature change. No weight influence is considered. If the pipe lifts off from the support due to temperature only, it is considered inactive for the expansion analysis.

This approach may mishandle both sustained and expansion stresses. First, the calculated sustained stress is the stress under ambient condition. The most important sustained stress

under hot condition is not calculated. Second, expansion stress may be underestimated, because the restraining effects of the supports, over which the piping is held down by weight, are ignored.

2. *Algebraic subtraction approach.* In this approach, sustained stress is calculated considering only the weight and pressure loads under ambient state. All supports are active because no temperature or support displacement is involved. An operating condition including temperature and weight is also analyzed, with the supports that lift off removed from the analysis. The expansion stress range is calculated by algebraically subtracting the hot operating condition by the weight condition at ambient state (initial cold condition).

Three major issues are at stake in this approach. First, the calculated sustained stress is the stress under ambient condition. The most important sustained stress under hot condition is not calculated. Second, by subtracting the initial cold weight load from the operating load, this approach includes the cyclic weight stress range, changing from ambient to hot conditions, in the expansion stress range. This is not consistent with the code philosophy of separating sustained stress from self-limiting expansion stress. Third, the stress for the temperature plus weight condition depends greatly on the signs of the moments of the two loads included. If the moment of the weight change is in the opposite direction of the moment of the temperature change, the calculated expansion stress will be smaller than that calculated by the temperature change alone. This is not correct because yielding and relaxation can change the sign of the thermal expansion moment during the course of operation.

Although the code allowable thermal expansion stress range includes the consideration of sustained stress, the combination of expansion stress range and sustained stress shall be done absolutely. Because the signs of expansion stress and sustained stress are immaterial, their algebraic summation or subtraction is meaningless.

3. *Operating condition approach.* In this approach, all supports are checked at the operating condition, which normally involves temperature plus weight and pressure. If the pipe lifts off from a support at operating condition, that particular support is then treated as inactive for both the sustained weight plus pressure stress calculation and the expansion stress calculation. By the same token, if the pipe is held down on a support at operating condition, that support is treated as active for both sustained and expansion stress calculations. With this method, sustained stress and expansion stress are calculated independently once the activity of the supports is determined.

The sustained weight plus pressure stress calculated with this approach is the true sustained stress at hot operating condition, when the stress is high and the pipe is weak. The one thing that may appear to be improper to some inexperienced analysts is that the calculated weight displacement may show a downward movement at support locations. This downward displacement represents only the movement of pipe from a thermally lifted condition. At support locations, the operational displacements combining weight and temperature will either be zero or in the upward direction. The calculated expansion stress is the potential stress range, recognizing that the sign and the magnitude may change throughout the operating cycles.

From the above discussions, it is obvious that the *operating condition approach* is the only method that meets the code philosophy and requirements. The other two approaches have flaws in calculating the sustained stress and the expansion stress range.

Analysis of piping resting on supports is nothing new. Engineers have routinely analyzed this type of piping for more than two decades. The erroneous concepts of some computerized approaches and the blind acceptance of computer results by engineers, however, are new. Attracted by the glamorous nature of thermal flexibility analysis, many engineers have forgotten that sustained stress is much more important than expansion stress. Sustained stress is the primary stress, whereas expansion stress is a

secondary stress. At low temperatures, when the hot allowable stress has the same value as the cold allowable stress, the weight allowable stress limit is only about one-third of the expansion stress allowable limit. At higher operating temperatures in the creep range, the weight allowable stress limit can be as low as only one-tenth of the expansion stress allowable limit. Therefore, it is important to note that the first priority of piping analysis is to accurately determine the sustained weight stress under hot operating condition. This is not to say that expansion stress is unimportant. A good analysis shall calculate both sustained and expansion stresses as accurately as possible.

It should be noted that by calculating weight stress at cold condition, the result is not expected to indicate where a spring support is needed. It is only when the weight stress at hot operating condition is calculated that the engineer will be able to detect where and when a spring support is needed. Spring supports are used to reduce the weight stress at hot operating condition.

The operating condition approach may be somewhat conservative for pipes that only lift up a very small amount from the support. In this case, the rule of thumb is to consider the support double-acting to check both sustained and expansion stresses. If both stresses are within the code allowable limits, then the system should be considered acceptable.

6.4 VARIABLE SPRING AND CONSTANT EFFORT SUPPORTS

As previously noted, the most economical and efficient way to support the piping is to simply rest the piping on a rigid support structure. Being supported with rigid supports, the pipe will either generate a potentially huge upward force when the pipe expands downward or leave the support inactive when the pipe moves upward. One way to maintain the pipe properly supported is to replace the rigid support with a spring support. With spring support, the pipe will always be supported with an appropriate amount of force, regardless of its vertical movement. The magnitude of the supporting force, however, changes somewhat as the pipe moves vertically. Because of this change in support force as the pipe moves up and down, spring support is also called a variable spring support. Although the support force changes throughout the operating cycle, the amount of change is predictable from the spring selected. In the design, the spring is properly selected so that the load variation is within the acceptable limit that will not compromise the integrity of the piping system.

Since the pipe always has the tendency to move in the vertical direction, it takes some engineering judgment to decide when and where the spring support shall be used. A straightforward method often mentioned is the use of the free vertical thermal movement of the piping as the criterion. In this method, the system is first analyzed for thermal expansion, without any support included, to find out the free vertical piping movement at the intended support location. If the free vertical thermal expansion exceeds a certain limit, say 0.5 in. (13 mm), then a spring support is used. If it exceeds 3 in. (75 mm), then a constant-effort support is used. This type of approach may appear to be reasonable, but actually does not serve any real purpose. At the location near a fixed point, such as an anchor or equipment, even a very small vertical displacement is too much to use a rigid support. On the other hand, a large vertical free thermal displacement in a wide-open area can still be supported with a rigid support due to the large flexibility of the piping. With a computerized environment, a good approach is to use double-acting rigid supports as much as possible for a trial. After the trial analysis, the supports that generate huge thermal loads are removed to see if the sustained stress is still within the allowable range. If it is not, then spring supports should be used at some or all of the locations where the trial double-acting rigid supports are eliminated.

In the configuration shown in Fig. 6.8, a 12-in. (300-mm) pipe will lift off and be unsupported at all three supports with a tower movement of 3.2 in. (80 mm). Ideally, all three supports should be replaced with spring hangers. However, with a compact layout as shown, only the mid-support may be required to be spring-supported, whereas the other two are simply removed. This, naturally, needs to be verified with an analysis. It should be also noted that some of the resting supports, although not active under operating condition, might be required for the hydrostatic test. However, a support

required for hydrostatic test but not suitable for proper piping operation should be removed after the test.

6.4.1 Variable Spring Hanger Selection Procedure

Spring hangers are selected based mainly on allowable load variation. The load variation is defined as

$$\text{Load variation} = \frac{\text{Cold load} - \text{Hot load}}{\text{Hot load}} = \frac{\text{Displacement} \times \text{Spring rate}}{\text{Hot load}} \quad (6.3)$$

Industry standards [5] limit the load variation to no more than 25%. This means that, regardless of the pipe movement, the spring will carry no more than 125% and no less than 75% of the properly balanced portion of load. The unbalanced load will be shifted to the neighboring supports or equipment. In actual applications, a smaller design load variation may be specified for critical piping systems, such as the ones operating at creep range or connected to rotating equipment.

In the past, due to the unavailability of long springs, it was customary to use a constant-effort support whenever the expected vertical movement exceeded 2 in. or 3 in. depending on the criticality of the service. Nowadays, as extra long spring hangers are widely available from vendors, piping engineers tend to use the variable spring support as much as possible due to its load adjustability. Constant-effort supports are reserved for extra large displacements that exceed the application range of the variable spring support.

For easier application in the spring hanger selection process, Eq. (6.3) can be rearranged around the spring rate as follows:

$$\text{Spring rate} = \frac{\text{Load variation} \times \text{Hot load}}{\text{Displacement}} \quad (6.4)$$

Equation (6.4) is the main engine of the spring hanger selection process, which is summarized into the following steps.

1. Determination of hot load

Hot load is the balanced load that exerts the least weight stress to the system. This load is obtained by analyzing the weight load with all supports considered as rigid. To minimize the load carried by the equipment, occasionally the vertical constraint at the equipment is mathematically released while the balanced load is being calculated. This balanced load is called the hot load because we want the spring to carry this load at operating condition.

2. Calculating the operational displacement

This operational vertical displacement is determined by analyzing the operating condition with weight and thermal expansion combined. The hot load determined in the previous step is applied as an external support force. Spring rates of the spring hangers are ignored, because the spring load applied is the hot load.

3. Selecting allowable load variation

The maximum allowable load variation is 0.25 (25%). Within this maximum variation, other limitations may be used depending on experiences and project specification [6]. A load variation of 0.15 (15%) is generally recommended for hot piping operating at creep range, and a load variation as low as 0.06 (6%) may be specified for piping connected to sensitive equipment.

4. Calculating the maximum spring rate

After determining the hot operating load, vertical pipe movement, and allowable load variation, the maximum spring rate is calculated using Eq. (6.4). A higher load variation results in a higher spring

rate. The actual spring rate used shall be less than the maximum spring rate calculated. The terms spring rate and spring constant are interchangeable.

5. Selecting the hanger supplier

Spring hangers are selected based on a supplier's catalog. Although a supplier can furnish almost any spring hanger specified, the cost is often prohibitive for specially manufactured hangers. Therefore, it is important to select the hanger that is offered in the supplier's catalog. In case the supplier has not yet been selected during the design analysis stage, a reference supplier's catalog can be used. One popular reference catalog is the Grinnell Pipe Hangers catalog [7]. You can, for instance, select the hanger with the Grinnell catalog, and then specify it as Grinnell Fig-xxx, Size-xxx, Type-xxx, or equal. This approach works, because most suppliers offer very similar products.

6. Selecting the basic spring

The supplier's catalog provides a selection chart similar to the one presented schematically in Table 6.2. Each supplier offers about 20 to 25 hanger sizes. The chart shows size 0 through size 22. Each hanger size has up to five displacement ranges: normal (N), short (S), long (L), extra long (XL), and double extra long (XXL). Suppliers have different names for these displacement ranges. Grinnell, for instance, calls them Fig-B268, Fig-82, and so forth. Because using the figure number to identify the displacement range has become a tradition, the displacement ranges are also referred to as "Figures." This practice also lessens the confusion with other types of ranges, such as the load range. Each hanger size also has its load range, the same for all figure numbers (displacement ranges). The size 2 hanger in Table 6.2, for instance, has a working load range of 95 lb to 162 lb regardless of the figure number. In addition to the working load range, there are also top and bottom margins to accommo-

TABLE 6.2
VARIABLE SPRING HANGER SELECTION CHART IN (LB),
(IN) UNITS - SCHEMATIC

		Range					Hanger Size					
		XXL	XL	L	N	S	0	1	2	*****	21	22
(Spring Displacement, in)	Top Margin	(0.0) (2.0)	(0.0) (1.5)	(0.0) (1.0)	(0.0) (0.5)	(0.0) (0.25)	43	63	81	*****	18750	25005
	Working Range, in	0	0	0	0	0	50	74	95	*****	21875	29173
							*	*	*		*	*
							*				*	*
							*				*	*
							*				*	*
							*	*	*		*	*
		10	7.5	5	2.5	1.25	88	126	162	*****	37500	50010
	Bottom Margin	(12) (14)	(9.0) (10.5)	(6.0) (7.0)	(3.0) (3.5)	(1.5) (1.75)	95	137	178	*****	40625	54178
							Spring Rate - (lb/in)					
						S	30	42	54	*****	12500	16670
						N	15	21	27		6250	8335
						L	7.5	10.5	13.5		3125	4167
						XL	5	7	9		2083	2778
						XXL	3.75	5.25	6.75	*****	1563	2084

Notes: 1 lb = 4.4482 N; 1 in = 25.4 mm; 1 lb/in = 0.175 N/mm

date the uncertainty involved with the data and calculation. Beyond the top and bottom margins, the hanger behaves just like a rigid hanger.

The hanger selection process starts with locating the hot load on the support load area to determine the hanger size. Up to two hanger sizes may contain a given hot load. For instance, a load of 110 lb is within the working load ranges of sizes 1 and 2. We shall select the smallest and shortest spring that is suitable for the operation. A smaller, shorter spring is cheaper and also requires less installation space, which can be very critical. Start from the smaller size, and look for the spring rate that is equal to or less than the maximum spring rate determined previously. From this spring rate, trace backward to find the corresponding displacement range, or figure number.

7. Checking the working range

The selected spring has to work within the working range. We already have the hot load located inside the working range; the next step is to determine if the cold load is also located inside the range. This cold load is calculated by:

$$\text{Cold load} = \text{Hot load} + (\text{Displacement} \times \text{Spring rate}) \quad (6.5)$$

In this equation, upward displacement is considered plus and downward displacement is minus. If the cold load is not within the working range, the other hanger size that also contains the hot load may be tried. If neither of the two hanger sizes works with the spring rate, a lower spring rate — that is, a figure with longer displacement range — can be tried. If no figure can accommodate the hot-cold load range, then a constant-effort support shall be used.

The spring hanger selection process is tedious and can be overwhelming for less-experienced engineers. The task, however, has been implemented in most pipe stress analysis computer software packages. The designer only has to instruct the computer where to put the hanger, what is the allowable load variation, and which supplier is to be used. The computer will automatically go through the procedure and print out a selection table as shown in Table 6.3. The table has all the information, except hanger type and overall dimension, needed to order the hanger.

6.4.2 Constant-Effort Supports

A constant-effort support is used when the load variation of a variable spring hanger exceeds its acceptable value or when the vertical pipe displacement exceeds the range of the variable spring hanger.

TABLE 6.3
COMPUTER GENERATED SPRING HANGER SELECTION TABLE

SIMPLEX REPORT PROCESSING										
File Reports Help										
SIMPLEX-PIPE STRESS ANALYSIS (7.0) (ASME B31.3) - PENG ENGINEERING, HOUSTON										

RESTING SUPPORT EXAMPLE										17 SEP 05
12 INCH SCH-40, 750F (400C), WITH 3.2-IN (80-MM) TOWER MOVEMENT										RestSpt.DA

*** SPRING HANGER SELECTION TABLE **										

USING -- GRINNELL -- VARIABLE SPRINGS										
(FOR TRAPEZE SPRINGS, THE LOAD SHOWN IS FOR EACH SPRING)										

POINT NO.	OPERATN LOAD (N)	COLD-SET LOAD (N)	VERTICAL-DISPL NORM (MM)	MAX (MM)	MIN (MM)	SPRING RATE (N/MM)	LOAD VAR (O/O)	GRINNELL-SPRG FIGURE	NUMBER SIZE	SPRING
30	11557	13437	53.7	53.7	.0	35.0	16.3	TRIPLE	13	1

A constant-effort support consists of a spring with an ingeniously proportioned linkage that offers a constant support force through a fairly large displacement range. Although constant-effort supports are mostly hung from above the pipe, they are traditionally called “supports,” leaving the term “hangers” to indicate variable springs.

Selection of the constant-effort support is very straightforward. The information needed is the operating load and travel range. Support vendors suggest that the specified travel range should be equal to the maximum actual operating displacement plus 20%, and in no case should the additional value be less than 1 in. (25 mm). The hanger size is then selected from the vendor catalog. One of the unique characteristics of the constant-effort support is the size-load relationship or, rather, non-relationship. The working load for the same size of support largely depends on the travel. This is somewhat puzzling, but can be easily explained.

Figure 6.9 shows the schematic representation of the constant-effort support. The support consists of a spring and a rocking yoke. The yoke rocks around a pivot pin located at the lower end of the support frame. One end of the yoke is connected to the pre-compressed spring, whereas the other end is connected to the load rod. The load, W , is supported by the pulling force, F , of the spring. By balancing the moment about the pivot axis, we have $FS = WL$. Because FS remains relatively constant for each size, the relation can be written as $WL = \text{constant}$ for each size of the support. From the sketch, it is obvious that the longer the travel, d , the wider the L required. Therefore, the greater the travel, the lower the support load that can be sustained by a given size of support.

Although the load of a constant support can be adjusted by about $\pm 10\%$ in the field, it does not offer any indication when the balanced load is reached. Because the load, once set, remains the same regardless of pipe position, the exact balance largely depends on an accurate calculation. To estimate the accurate support load, some potential variations on the weight of pipes and components have to be considered. An ordered pipe normally has some overweight. The overweight is even more common for components such as forged elbows and tees. The weight of pipe clamps and other suspended support components have to be included as concentrated weight for the calculation to be accurate.

6.4.3 Spring Support Types and Installations

The computer-generated spring hanger selection table does not give the support type, which is dictated by the availability of space and support structure. Each spring hanger supplier normally offers about seven or eight different spring hanger types for installing in different situations. Figure 6.10 shows typical applications of hanger types offered by Grinnell Corporation [7].

Regarding the installation of spring hangers, different practices have evolved for different industries. Power plant piping is typically hung from the floor structure above, using rigid hangers. When a spring

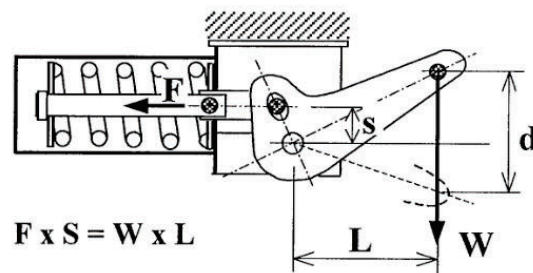


FIG. 6.9
CONSTANT-EFFORT SUPPORT SCHEMATIC SKETCH

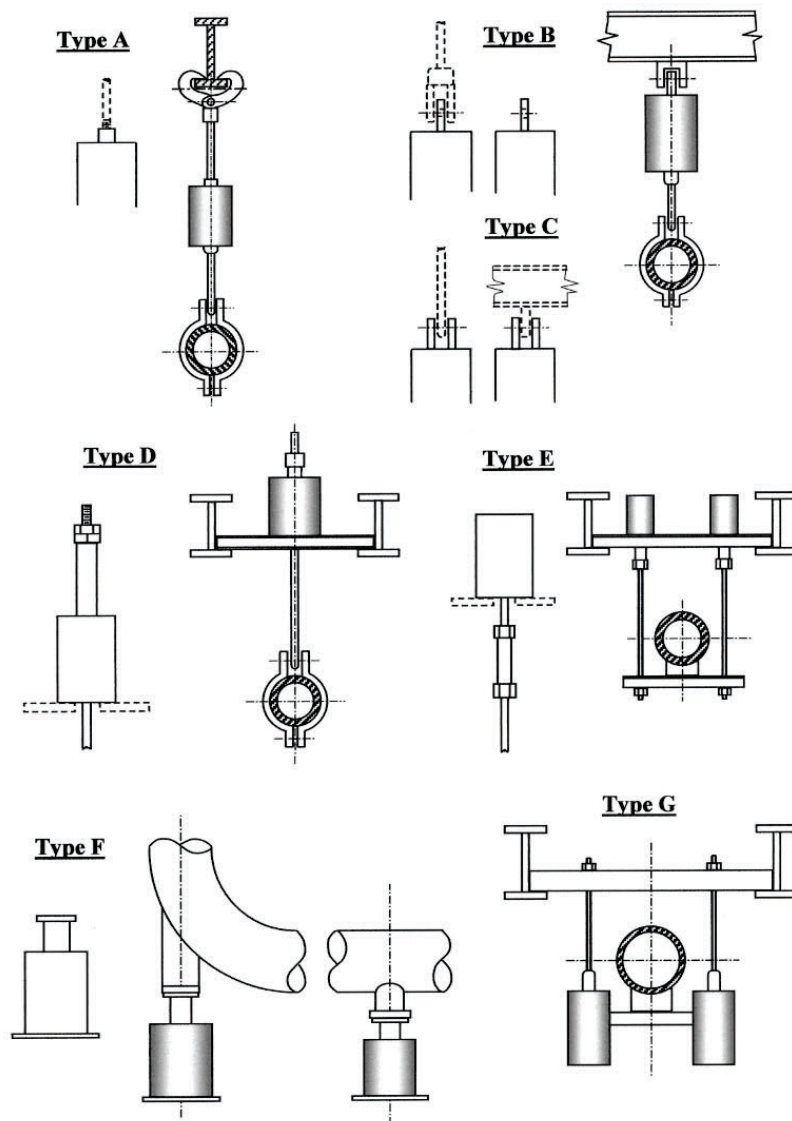


FIG. 6.10
SPRING HANGER TYPES AND APPLICATIONS (BASED ON GRINNELL SPRING [7])

hanger is needed, it is just a matter of replacing the rigid hanger with a spring hanger. The headroom is generally not a problem. In this case, the piping layout seldom requires any change because of the use of spring hanger. On the other hand, process piping is normally resting on the support structure underneath, without any support structure available overhead. In this case, the use of spring hangers

or spring supports is an important matter. In many occasions, the piping layout has to be revised to accommodate not only the spring hanger hardware, but also the piping movement. In some process plant projects, the use of spring hanger and sometimes even rigid rod hanger requires prior approval from the project management.

6.4.4 Setting of Loads — Hot Balance and Cold Balance

The load of a spring hanger changes throughout the operating cycle. By setting the pre-operation cold condition load at a certain level, we can predict the load at operating condition based on the expected vertical displacement. Again, load variation is calculated as spring rate times the vertical displacement. The load that balances the weight of the portion of the piping supported is called the balanced load. Intuitively, we might like to set the balanced load at midway of the operating cycle and shift one-half of the load variation to the cold condition and the other half to the hot condition. However, in actual applications, the piping system is either hot balanced or cold balanced.

Hot balance sets the hanger in such a way that the spring load at hot operating condition equals the balanced load. This is the preferred approach that, at least theoretically, minimizes load and sustained stress at the most important hot condition. This is a must for piping operating at creep range, where creep damage depends largely on sustained stress at operating condition. Figure 6.11 shows the scheme of setting the spring given in Table 6.3. The hot load is equal to the balanced load of 11,560 N (2600 lb). The setting of the cold load is then determined by adding the expected load variation to the hot load — that is, the cold load is 13,440 N (3020 lb). The spring is locked at this cold load with a red-painted locking lug or pin. A colored warning label is attached to remind the operator to pull out the pin before operation. After the installation and hydro test is completed, the locking lug or pin is removed. Because at cold condition the hanger is pulling more force, or less force as the case may be, than the balanced load, the piping will experience some jerk or movement when the lug is removed. This can alarm field engineers if the movement is significant. Fortunately, the lugs are removed one spring at a time so a very large movement is rarely observed. If all calculations are exact, the magnitude of the unbalanced load is within the allowable load variation with which the system is designed for. For the spring shown in Fig. 6.11, the spring force reduces gradually as the pipe temperature rises and the piping moves up. The spring force eventually reaches the balanced load of 11,560 N when the piping system reaches the operating temperature.

Cold balance sets the hanger force to balance the weight load at cold condition. This leads to some unbalanced force at operating condition when the piping is at hot operating condition. In theory, this approach is not as good as the hot balance option. However, most experienced field engineers, especially those who represent large machineries operating at low to moderate temperature, insist on using cold balance on the piping systems attached to their equipment. This often creates arguments between the machinery engineer and the piping stress engineer. The personnel friction can be avoided and the job can be better handled if we know the rationale behind the equipment engineer's insistence.

Accurate support loads can only be determined from accurate pipe data and calculations. The mathematical portion of the equation is not a problem when using a computer program, but the data portion is much less reliable. The pipe weight can be 10% heavier than the theoretical value, and the forged fitting can be as much as 30% heavier than the theoretical value calculated from the equivalent pipe. The insulation covering and support clamps and floating attachments are often not included in the calculation. Because some of these items are not known for sure, the calculated support load is naturally less than exact. Furthermore, the calculation model often ignores connecting piping of smaller sizes, and the boundary conditions used are just approximate assumptions. Therefore, it is possible that the cold set load is way off the presumed value, and the resulting hot load is not at all predictable.

The displacement of the low to moderate temperature piping, such as some of the large compressor piping systems at a process plant, is generally small. This means that the actual load variation,

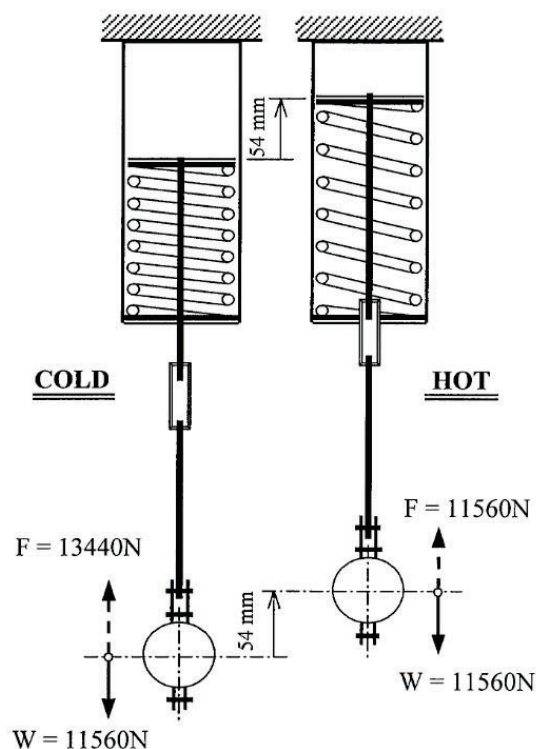


FIG. 6.11
HOT BALANCE SPRING LOAD

calculated as the product of spring rate and displacement, and the percentage of load variation are also generally small. Therefore, if there is a means to achieve the exact balance at cold condition, then the unbalanced load at hot condition is small and predictable. The method used by the equipment engineer is to apply no load at all on the connecting flange or the final field weld when the piping is being connected to the equipment. The springs are unlocked after the hydro test. The piping is then pulled using the support, spring or rigid, to bring the connecting flange to the equipment matching flange without the use of any other external forces. In this way, it is fairly certain that the weight load on the equipment at cold condition is practically zero. The hot load, which results from spring load variation, is small and predictable. This approach of connecting the piping to equipment, however, is only feasible for gas or vapor piping.

Cold balance is very popular in the field, but somehow does not yet capture the attention of stress engineers. To facilitate the field installation process, the spring should be set in such a manner that the cold load is in balance with the weight of the portion of the piping being supported. The vertical restraint at the equipment connection should be released mathematically when calculating the spring load, so the potential weight load at the equipment connection is as small as possible. To achieve the no-load connection, the supports have to be well distributed and positioned throughout the piping system. Without the proper pulling points, the no-load situation can never be achieved.

6.5 SUPPORT OF LONG RISERS

A long vertical line is called either a riser or a down comer, but is generally referred to as a riser. A riser has a few unique characteristics that require special attention. The riser has the potential of producing a large vertical expansion, has a large concentrated weight acting at one point in the horizontal plane, and has a large liquid column if it is a liquid line. These unique characteristics affect the support scheme and analytical process. In the following, we will discuss the support scheme, load calculation, and analytical approach for the risers.

Figure 6.12 shows three typical cases of riser arrangements. In case (a), the riser is hung directly underneath a piece of equipment such as drum or vessel; case (b) shows a riser connected to two considerably long horizontal runs at both top and bottom; and case (c) shows a typical process line running alongside a tall vessel.

6.5.1 Support Schemes

Because of the large vertical displacement, it is a natural tendency for engineers to use constant-effort or long variable spring supports. However, due to the large concentrated weight, a slight amount of load variation or inaccuracy of data leaves a significantly unbalanced weight that can generate a large bending moment at the horizontal portion of the piping. A rigid support is required somewhere nearby to absorb this load variation and uncertainty. In case (a), the entire riser grows down at hot operating condition. There is no choice but to put the spring supports as close to the equipment as possible to minimize the vertical displacement. In case (b), there is always a theoretical zero vertical displacement point located at the riser. Rigid supports or hangers located at close proximity of this

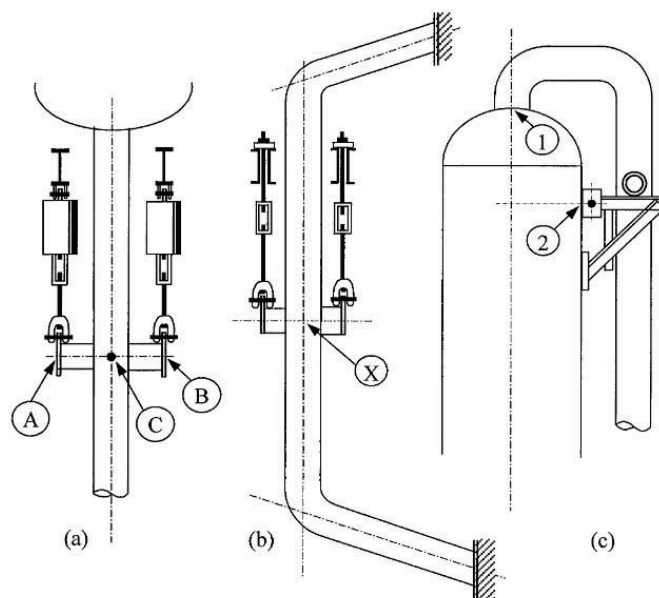


FIG. 6.12
SUPPORTS AT RISERS

point should be used. Rigid hangers are generally more convenient as they can be hung from overhead structural steel. Due to the flexibility of the horizontal runs, the location of the zero vertical displacement point needs not be exact. Case (c) is a typical process line connected to a tall vessel. Because the temperature of the part of the vessel at the connection is the same as the pipe temperature, assuming the vessel is not internally insulated, a rigid support can always be used and located close to the connection. For a very hot long riser, it may require supports at multi-levels to reduce the weight stress.

6.5.2 Support Loads

Support loads can be determined by allocating the proper portion of the piping system, including pipes, components, attachments, liquid content, and insulation, to each support. This is done by a mixture of common sense and simple arithmetic. In a computerized design environment, the computer automatically determines the loads. Computerized calculation is quick and accurate, but the determined loads are often neither what we want nor are they the most suitable ones. Several special procedures have to be applied in performing the computerized load determination analysis.

In case (a), a normal computerized analysis will allocate most of the weight to the equipment, as it is the most rigid support. Some spring load is assigned to one of the springs, leaving another spring with no load at all. This, of course, is not what we want. What we want is to have the spring carry most of the weight and to have the weight shared equally by the two springs. To achieve this, first, the vertical translational restraining effect at the equipment has to be released when calculating the weight support load. This anchor release procedure will leave essentially no weight load at the anchor or equipment. Most computer software packages will automatically do this if instructed. To distribute the spring load, a seemingly more accurate analysis scheme is to model trunnions A and B as two short pipes, each connected with a spring. However, this elaborate approach often produces two completely different springs with occasionally no load at all at one of the springs. This is attributed to the theoretical approach of the computer software. When the computer distributes the weight support load, it considers all spring supports as rigid in the calculation. This works just fine if the supports are well separated. With two rigid supports located close by, as in this case, only one support will be active due to the cranking action of the horizontal run located down below. The proper method of analysis is to consider the effect of the two springs as a combined spring located at the center point, C, and then determine the support load at this center point. The load so determined is divided by 2, from which the load of each spring is obtained. Because the two spring loads generate zero combined moment at the center point, the one center spring approach is mathematically equivalent to the two separated, but equal, springs. The trunnions have to be modeled, however, to have their weight counted. The weights of hanger attachments are lumped in the analysis as concentrated weight acting at the center point. In most pipe stress analysis software packages, the two springs can be automatically selected by instructing the computer to select two springs at the center point.

The loads on the rigid hangers in case (b) are difficult to determine both analytically and practically. A computerized analysis will most likely predict that only one hanger will carry the weight load due to the cranking action of the horizontal runs. The other hanger will simply sit there with no load. The roles of the two hangers, however, might be switched when the system is at operating condition. Although the loads on rigid hangers can be adjusted or shifted somewhat by tightening the rods, they are not predictable. Therefore, unlike in case (a), the method of using one-half of the load determined at the center point is not feasible. The computer-calculated loads for the rigid hangers have to be used. This means that the total load is generally carried by only one of the hangers. More realistic loads may be calculated by including the proper stiffness or spring constant of the hanger and support structure in the analysis. The stiffness of the rigid hanger and structure assembly is in the range of 10^5 lb/in. to 10^6 lb/in. ($1 \text{ lb/in.} = 0.175 \text{ N/mm}$).

The situation for case (c) is similar to that of case (b), but with some differences. Here, the riser is generally long and slender and offers some flexibility to seat both sides of the support. The support

structure is a combined assembly used for both sides of the support. In general, if the support trunnions are shimmed properly, then the theoretical support situation can be achieved. Even so, some extra margin needs to be provided for the trunnion design load.

6.5.3 Analysis Method

In the analysis, insulation and content are lumped together with the pipe weight. In other words, the analysis is based on total weight, including pipe, insulation, refractory, and content, per unit length of the pipe. That is, a unit length of the pipe is combined with a unit length of insulation and a unit length of liquid to become the total weight per unit length of pipe in the analysis. This may appear to contradict the fact that the liquid weight of the entire riser acts upon the bottom of the riser, rather than on each unit length of the pipe. This convenient approach, however, does not affect the support load distribution calculation, as all weights are still acting on the same vertical centerline. As for the sustained stress calculation, the method is conservative as soon as the design pressure is taken as the pressure existing at the bottom of the riser.

6.6 SIGNIFICANCE OF SUPPORT FRICTION

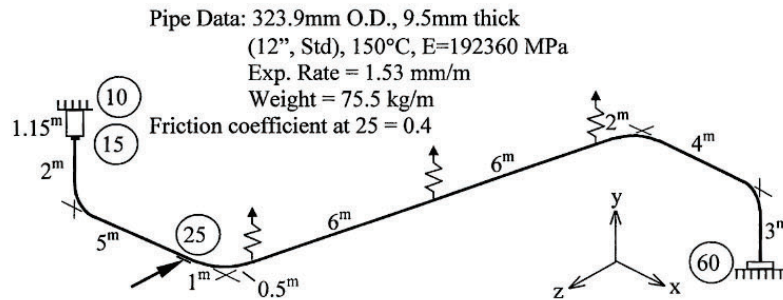
A piping system may involve several different types of friction force. Flexible joints, such as Dresser couplings and ball joints, need a certain breakup force to start sliding or rotating. These required breakup forces are due to the friction of the tightly packed joint. These friction forces will be discussed in Chapter 7. Another type of friction involving soil pressure will be discussed in Chapter 10. The friction forces discussed here are the ones produced at the sliding surfaces of the supports.

6.6.1 Effects of Support Friction

Support friction in a piping system prevents the pipe from thermal expansion, thus creating higher stress in the piping and higher load on the connecting equipment. However, in certain instances friction can help stabilize the system and reduce potential damage. Even in dealing with pure thermal expansion, friction can serve as guides preventing a large load from transmitting to a piece of sensitive equipment. Therefore, there is no rule of thumb as to whether it is non-conservative to ignore friction. In general, when dealing with dynamic loads, friction tends to reduce the magnitude of both the pipe stress and the equipment load. In this case, the omission of friction in the design analysis is conservative. However, there is no general rule governing the static load. With static loads, the effect of friction needs to be investigated and simulated as closely as possible to the real situation. When designing the support structure, however, it is always required to include the pipe support friction forces.

The effect of support friction is very important in some areas. Analysis of the long transmission pipeline, for instance, focuses entirely on the balancing of friction force against potential expansion force. If friction is not included in the analysis, the analysis would be meaningless. Another area of importance is the piping connected to rotating and other sensitive equipment. Rotating equipment is notorious for its low allowable piping load. Sometimes the friction at one support can completely change the acceptability of the piping system. Take the system shown in Fig. 6.13, for instance, where the restraint at point 25 is installed to protect the compressor at point 10. The effect of friction at point 25 is demonstrated by comparing the analysis results [8] of the case with friction against the case without friction. It is clear that the effect of friction at the restraint is very significant. By applying the American Petroleum Institute Std-617 [9] criteria, only the load calculated with the restraint but without friction is acceptable. Restraint friction has to be either removed or substantially reduced (see also Chapter 9).

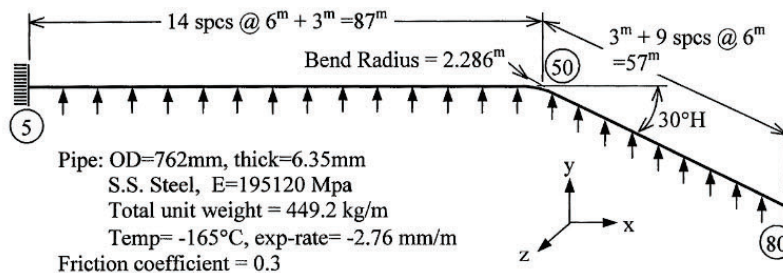
The effect of support friction is also very significant for long off-site piping. Figure 6.14 shows a zigzag section of liquefied natural gas piping. From the comparison chart, it is clear that support



Piping Loads at Compressor Nozzle 15						
Conditions	Forces at 10, (N)			Moments at 10, (N-m)		
	F _x	F _y	F _z	M _x	M _y	M _z
No Restraint	-1145	-3727	3227	-4518	-10152	-1829
Restraint without friction	-1485	-3913	-289	1143	-17	-2988
Restraint with friction	-4735	-3397	-339	-591	206	-7308

FIG. 6.13
EFFECT OF SUPPORT FRICTION ON COMPRESSOR PIPING

friction dominates the response of the system. Without support friction, the axial anchor force stems mainly from the stiffness of the system resisting the thermal expansion. Because the system is very flexible, this thermal resisting force is very small at 16,746 N as compared with the frictional resistance of 222,600 N. The support friction tends to prevent the piping from moving to the flexible direction,



Comparison of Analysis Results			
	Anchor Load at 5		Bend Moment at 50
	F _x (N)	M _y (N-m)	M _y (N-m)
With Friction	222600	10000	328380
Without Friction	16740	123942	124462

FIG. 6.14
EFFECT OF SUPPORT FRICTION ON OFF-SITE PIPING

because it only takes a very small force to stop the movement in the flexible lateral direction. Therefore, support friction also serves as a guide to a long flexible line. This is the reason why the moment at anchor 5 is much smaller with support friction than the moment for the case with support friction ignored. In addition, because of the guide effect, support friction tends to make the system less flexible. This causes the bending moment, thus the stress, at the bend to be almost three times as high when support friction is present as compared with the case when support friction is ignored.

6.6.2 Method of Including Friction in the Analysis

There are a few general steps to follow when including support friction in the analysis. The nature of friction is first idealized as elastic perfect plastic resistance as shown in Fig. 6.15(a). When the pipe is forced to slide on the support surface, initially it does not move because of friction resistance. As the pushing effort increases to a certain limit, the pipe starts to move and follows with a slight snap because of the drop in friction resistance after the pipe has started moving. The initial resistance force is called static friction, and the resistance after the pipe moves is called sliding friction or dynamic friction. Because friction at each support reaches the maximum level of static friction at different times, the lesser dynamic friction is used in the design of piping system. The model of friction force is, therefore, similar to a linear elastic/perfect plastic non-linear restraint. The linear elastic portion represents the flexibility of the support. This linear portion is also needed for stabilizing the mathematical simulation processes.

The analysis is an iteration process involving a trial-and-error approach. There are two main approaches used in piping stress analysis: direct friction force approach and resilient friction restraint approach.

The *direct friction force approach* starts the analysis by first assuming that friction stops the pipe from sliding on the support surface. The resultant stopping force is then calculated to compare with the full friction force, $F\mu$. If the resultant stopping force is less than the full friction force, then the pipe is stopped, as assumed. If the resultant stop force is greater than the full friction force, then

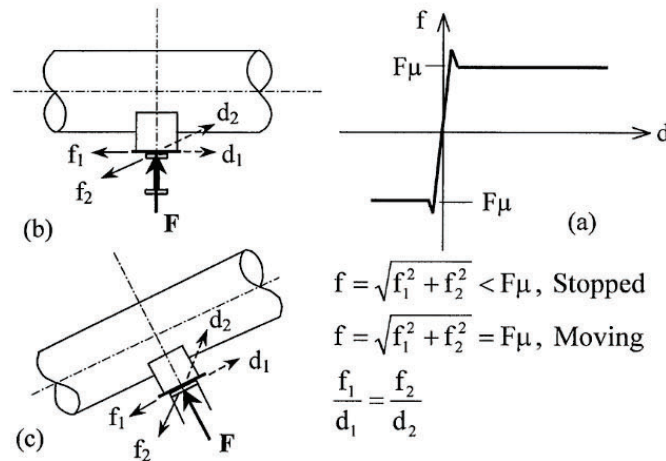


FIG. 6.15
SUPPORT FRICTION COMPONENTS

the pipe slides with a total resistance force equal to the full friction force. The analysis is performed again, assuming that there is no stop to pipe sliding, except the full friction resistance force, at those support locations where the stop force is bigger than the friction force. In the analysis, friction force is applied as external force in the direction against the pipe movement. Because the sliding of the pipe on the support surface involves two degrees of freedom, the friction force is two-dimensional. It comprises two perpendicular forces applied in the direction perpendicular to the support direction. Figure 6.15(b) and (c) shows the schematic directions. As the friction force works directly against the displacement, its components, f_1 and f_2 , are proportional to the displacement components, d_1 , and d_2 , as shown in the figure.

The general analysis procedure, described above, works well for some simple and rather stiff systems, such as the compressor piping given in Fig. 6.13. In this system, the reaction force at point 25 is first calculated without the friction. Based on the calculated restraint reaction force, a full friction force is determined and allocated proportionally to the displacement components in x and y directions. The calculated friction components f_x and f_y are then applied as external forces in a re-analysis. The restraint reaction and pipe displacements will likely change after the re-analysis. A new set of adjusted friction components are again applied as external forces in another trial analysis. Because the system is rather stiff, a convergence will be reached in just a couple of trials. However, this direct friction force as external force approach does not work very well on long flexible piping systems involving many supports.

The *resilient friction restraint approach* is more stable especially for long and flexible systems, such as the one shown in Fig. 6.14, in which the direct friction force approach does not work at all. With the direct friction approach, the trial lateral friction forces can twist the system into a very unpredictable situation, and the trial-and-error iteration can deviate farther and farther away from the balanced position. One workable scheme to deal with this type of flexible system is to model all potential friction forces as resilient restraints. Each component of friction force is represented by a resilient restraint. A resting support in y direction, for instance, will have two resilient friction restraints in x and z directions, respectively. In each trial analysis, the spring rates of these resilient restraints are determined from the potential friction forces and the pipe displacement of the previous trial analysis. Because the simulated friction force is resilient, no extreme pipe displacement will be produced. The analysis generally converges to the balanced state in a few iterations, although some systems may take a couple dozen iterations to reach the balanced state. The resilient restraint approach is summarized as follows:

- (1) Each friction support and restraint is assigned with two orthogonal resilient friction restraints perpendicular to the direction of the support or restraint. The resilient restraints are aligned with the local coordinates of the sliding surface, and their reaction forces represent the expected friction forces.
- (2) The analysis starts out with no friction effect to calculate the preliminary support reaction force, F , for each support or restraint. It also calculates the preliminary displacements at two perpendicular directions to the support or restraint. The two perpendicular directions, of course, coincide with the directions of the resilient friction restraints. For a resting support in the y direction, for instance, the two displacements needed are d_x and d_z .
- (3) Each iteration of the analysis is performed using the updated trial spring rates for the resilient friction restraints. The spring rate is updated with current support reaction force and displacement information as

$$\text{Spring rate} = \frac{F\mu}{\sqrt{d_1^2 + d_2^2}} = \frac{F\mu}{\sqrt{d_x^2 + d_z^2}}$$

The same spring rate is used for both x and y directions at each support or restraint, so the direction of the resultant friction force is directly against the displacement direction.

- (4) The results of each iteration are checked to see if variations from the last iteration are all within the range of acceptable values. If the variation between current iteration and last iteration is not acceptable for any support, the new spring rates are calculated and a new iteration is performed.

6.6.3 Application of Friction Force

The procedure outlined is for the analysis including all loads, such as weight, thermal, etc. This is the condition required for evaluating equipment loads. However, in code stress compliance analysis, it is often required to separate the loads into different analyses so stresses can be separated into proper categories. For instance, the weight load is not included when the thermal expansion stress is being calculated. However, without the weight load, friction is mostly non-existent. This becomes a paradox in piping stress analysis, because friction would never get into the act. Friction is irrelevant in the weight load case, yet non-existent in the thermal expansion case.

Weight initial load is the key element for solving the above paradox. The cited inconsequential friction paradox can be solved by maintaining the initial weight load for each support or restraint. An analysis considering only the weight load is performed to determine the magnitude of the weight load that is normally carried by each support or restraint. These nominal weight loads are then recorded and remembered. We will then recognize that these weight initial loads are always present at the supports to affect the friction effect. The friction force can then be calculated with this weight initial load superimposed on the support reaction force produced from thermal expansion and other loads.

Because the friction is mostly the result of weight load, some may argue that friction force is a sustained load simply because weight is a sustained load. This, of course, is not true because friction force is a passive force. It is always in the defensive mode (and is never in an active mode to affect anything). Friction force will tend to reduce the effect of a sustained load, but it does not have the damaging nature of a sustained load. Friction is actually an effect rather than a force.

Friction generally serves as restraints to a piping system. It increases the effective stiffness of the system. Obviously, the stiffer the system, the higher the load, and also the higher the stress, which will result in the piping acting against thermal expansion. Therefore, inclusion of friction in thermal expansion analysis is necessary. On the other hand, friction can help reduce the damage from occasional load, such as earthquake and wind. In these cases, the inclusion of friction is a matter of judgment. It should be clearly specified in the design specification.

6.6.4 Methods of Reducing Friction Force

Support friction is not desirable in some situations. In the examples given in Figs. 6.13 and 6.14, friction has increased anchor reactions and pipe stresses. The necessity of reducing support friction is seldom the consequence of pipe stress. The reduction is mostly required for reducing the equipment load and support structure load. We are all aware of just how small the allowable piping load can be at an equipment connection. However, we often forget that the support structure cannot take a large friction force either. Most support structures are mainly designed for the vertical load. They are generally very weak in resisting the horizontal load.

In the system shown in Fig. 6.13, restraint 25 generates a high *x*-direction friction force, causing a very large *x*-direction equipment force and *z*-direction equipment moment. Because these loads are not acceptable, one way to solve the problem is to reduce the friction force.

The best way to reduce support and restraint friction is to eliminate the sliding surface. This can be achieved by using hangers and struts. The strut as shown in Fig. 6.4(a) can be used here. However, there are situations in the field wherein the use of hangers and struts is not feasible. For example, for off-site and yard piping in a process plant, sliding support is often the only feasible support.

There are two popular means of reducing friction force at support and restraint surfaces: rollers and low-friction sliding plates. Figure 6.16(a) shows a sketch of the roller support, which converts sliding action into a rolling action. Roller supports not only offer less friction than the sliding support, but also help maintain the alignment of the pipe. They serve as guides for the pipe and, at the same time, accommodate slight axial rotation of the pipe. Rollers are popular for piping that is laid in a trench or in a dusty environment.

Figure 6.16(b) shows the general arrangement of low-friction sliding plates. Two major types of materials are used for the sliding plate: graphite and Teflon (poly-tetra-fluoro-ethylene, PTFE). Teflon is used at lower temperatures, from -330°F to 300°F (-200°C to 150°C), whereas graphite is used mainly at higher temperatures. Teflon is often filled with glass and other agents to increase the allowable bearing pressure. The sliding pair assembly may either have two low-friction plates or have one low-friction plate paired with a polished stainless steel plate. Both cases have about the same coefficient of friction, although the one with the stainless steel plate has a somewhat higher coefficient. The assembly with one low-friction plate paired with a stainless steel plate is more popular due to its sturdy construction and resistance to tearing off.

The friction coefficient of the Teflon sliding plate is very consistent at about 0.05. However, at contact pressures less than 300 psi (2000 kPa), the coefficient is high and unpredictable. Therefore, it is important to maintain the contact pressure at above 300 psi. The general guideline is to maintain the pressure at a minimum of 500 psi (3500 kPa) with a 0.1 design friction coefficient. In addition, the allowable bearing pressure of the Teflon sliding plate is rather low at the 2000 psi range. At 100°F , for instance, the allowable pressure is 2000 psi for an unfilled Teflon plate and 4000 psi for a filled Teflon plate. The allowable pressure decreases rapidly with an increase in temperature. These limitations imply the need for a rather precise contact area for each support. The use of the standard size plate is not permitted in many cases. The size of the contact area is likely to change when the pipe moves. One

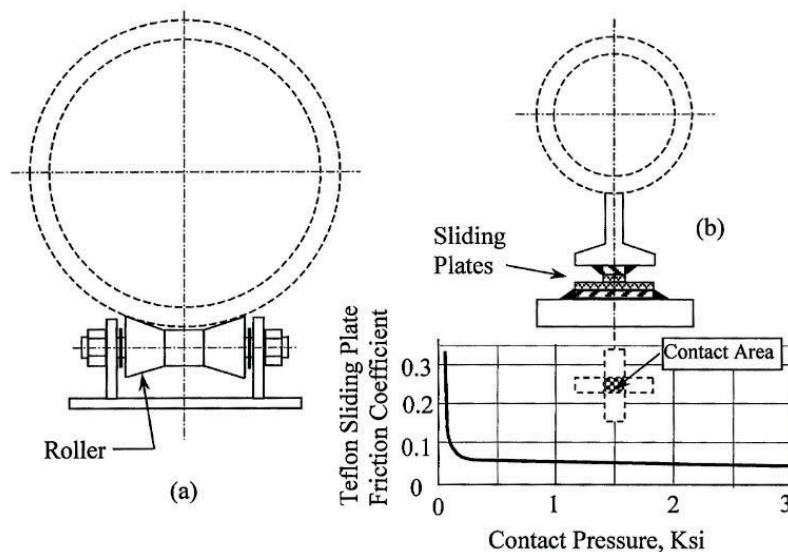


FIG. 6.16
LOW FRICTION SUPPORTS

way to ensure the size of the contact area is to use two perpendicular narrow rectangular strips, one on each side of the contact surface.

There are some difficulties associated with pipe support sliding plates. Because of the small size used in low load situations, the plate often drops off the contact surface when pipe movement is significant. Moreover, because sliding plates are required to be perfectly parallel to the surfaces to work, special installation procedures are required. Slight rotations of the pipe can also substantially reduce the effectiveness of the plates. Overall, low-friction sliding plates have been shown to be less than ideal in piping with large displacements. Movement should be minimized as much as possible by providing more frequent loops and offsets.

6.7 SUPPORT OF LARGE PIPES

Large pipes pose some special requirements in their support and analysis technique, mainly due to their high r/t ratio. Engineers tend to consider pipes larger than 24 in. (600 mm) in nominal diameter as large pipes, because some piping components standards cover only up to 24 in. However, the need for special requirements is not very clear-cut. It needs to be investigated on a case-to-case basis.

Large pipes generally tend to have higher r/t ratio due to the generally smaller internal pressure and smaller percentage of surplus thickness. The pipe shell of a large pipe has a higher tendency to deform at support locations than that of a smaller pipe. Once the shell deforms locally, the stress field becomes very complicated. Therefore, the current design procedures on large pipe depend mostly on field experiences and empirical data.

The load of the pipe, weight or otherwise, is carried by the shell with the tangential shear force distributed by a sinusoidal function as shown in Fig. 6.17. The tangential shear force per unit length, V , at location ϕ degree away from the zenith (top) is

$$V = \frac{W}{\pi r} \sin \phi \quad (6.6)$$

where W is the total shear force acting at the given cross-section under consideration. The equation shows the tangential shear force distribution at an undisturbed area without any support or restraint located nearby. In this case, the maximum shear force, which is located at the equator, is equal to $W/\pi r$. This maximum value is twice as big as the average value of $W/2\pi r$. This doubling of the average shear force and thus the shear stress is called the shear distribution factor as noted in Chapter 2.

As the shear force transmits to the support location, its distribution gradually changes to a very complex pattern at the support point. However, according to Saint-Venant's principle [10], the distri-

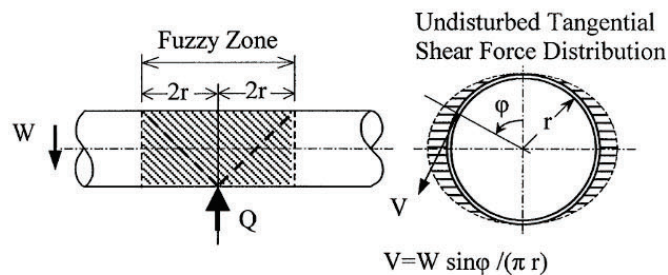


FIG. 6.17
TANGENTIAL SHEAR FORCE DISTRIBUTION